



BACKEND NODE.JS

Session 3



Date: 9-12-2025

Instructor: Tahany Emad





Outlines

- Introduction to JavaScript
- Data Types (primitive, non-primitive)
- types of variables (Var, let const)
- Operators
- Numbers Object
- Strings Object
- Math Object
- Conditionals
- Arrays
- Loops



Why JavaScript

- JavaScript Can Change HTML Content.
- JavaScript Can Change HTML Styles (CSS)
- JavaScript Can Hide and Show HTML Elements

JavaScript Where To

- In HTML, JavaScript code is inserted between `<script>` and `</script>` tags.
- JavaScript in `<head>` or `<body>` (Interpreter or Compiler)
- External JavaScript
- places we can write JS code (window, Browser, Console)



JavaScript Variables & Identifiers

- What is a Variable?

A variable is a container that stores data in memory so we can use it later.

Keyword	Can be updated?	Can be redeclared?	Scope
var	✓	✓	Function Scope
let	✓	✗	Block Scope
const	✗	✗	Block Scope

Note:

- var cares only about functions
- let and const care about blocks

- What is an Identifier?

An identifier is the name given to a variable, function, class, etc.



Rules for Identifiers

- Must start with a letter, `\_`, or `\$`
- Cannot start with a number
- Cannot contain spaces
- Cannot use reserved keywords (like `if`, `class`, `return`)
- Case sensitive → `name` ≠ `Name`



JavaScript Data Types

Type	Mutable / Immutable	Copying method	Examples
Primitive	Immutable	Copy by value	string, Numbers, Boolean, null, undefined
Non-Primitive	Mutable	Copy by reference	object, array, function

Primitive → cannot change → create new copy

Non-Primitive → can change → affects all reference



Operators

- Arithmetic Operators(+ , - , *, / , **, %)
- Assignment Operators(*=, /=, -=, +=)
- Increment / Decrement(++ , --)
- Comparison Operators(==, ==, !=, !=, >, <, >=, <=)
- Logical Operators(&&, ||, !)





JavaScript Numbers

- JavaScript will try to convert strings to numbers in all numeric operations except for **addition (+)**.
- If one of the operands is a string in a + operation, JavaScript will perform string **concatenation** instead of arithmetic addition

NaN - Not a Number

- Result of invalid math
- NaN doesn't equal itself because it represents an "invalid number".
If the value is unknown or not a real number





Number Method

- **toString()** : Converts number to a string.
- **toFixed(x)**: Formats number with fixed number of decimals
- **isNaN(x)**: Checks if value is the special NaN
- **parseInt(x)**: Converts string to integer (stops at first non-digit).
- **parseFloat(x)**: Converts string to float preserving decimals
- **isInteger(x)**: Checks if value is an integer.



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